

Civilization-State Asset Pricing

A Unified Theory of Structural Economics, oliGARCHy Dynamics, and Stochastic Discount Factor Reconstruction

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Abstract

This paper develops a unified framework integrating Structural Economics, oliGARCHy dynamics, and stochastic discount factor (SDF) theory into a single latent-state model of civilization. We argue that financial markets, political systems, geopolitical competition, technological development, and military power may all be interpreted as observable projections of a hidden civilization state. The stochastic discount factor is elevated from a pricing kernel to a state variable governing intertemporal valuation. Asset pricing, macroeconomics, political economy, international relations, warfare economics, and machine learning are unified through a common inverse problem: inferring latent civilization dynamics from observable market prices and strategic behavior. The resulting framework extends traditional asset pricing toward a general theory of valuation and civilization-scale state estimation.

The paper ends with “The End”

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1 Introduction

Modern finance traditionally studies the valuation of securities.

The central equation is

$$P_t = \mathbb{E}_t[M_{t,T}X_T],$$

where

- P_t is price,
- X_T is future payoff,
- $M_{t,T}$ is the stochastic discount factor.

Conventional theory treats the SDF as a pricing object.

This paper adopts a different view.

We interpret the SDF as one coordinate of a broader civilization state.

The objective becomes:

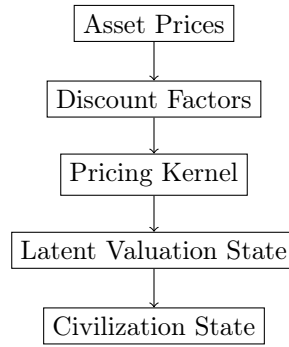
Infer the hidden civilization state from observable markets.

2 The Valuation Hierarchy

We begin with the hierarchy

Observed Prices $\implies$ Discount Factors $\implies$ SDF $\implies$ Civilization State.

The progression is illustrated below.



3 The Civilization State

Definition 1. *The civilization state is the vector*

$$X_t = \begin{pmatrix} S_t \\ V_t \\ M_t \end{pmatrix},$$

where

- S_t is the structural state,
- V_t is the volatility state,
- M_t is the valuation state.

4 Structural Economics Layer

Define

$$S_t = \begin{pmatrix} K_t \\ L_t \\ A_t \\ I_t \\ G_t \end{pmatrix}.$$

Components are:

Variable	Interpretation
K_t	Capital stock
L_t	Labor
A_t	Technology
I_t	Institutional quality
G_t	Geopolitical position

Output is

$$Y_t = I_t A_t K_t^\alpha L_t^{1-\alpha}.$$

Technology evolves according to

$$A_{t+1} = A_t + \eta_A R_t + \varepsilon_t^A.$$

Institutional dynamics satisfy

$$I_{t+1} = I_t + \eta_I \Phi_t + \varepsilon_t^I,$$

where Φ_t denotes institutional adaptation.

5 oliGARCHy Layer

Define

$$V_t = \begin{pmatrix} \sigma_t \\ \omega_t \\ c_t \end{pmatrix}.$$

Interpretation:

Variable	Interpretation
σ_t	Volatility
ω_t	Concentration of influence
c_t	Contagion intensity

Volatility evolves according to

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \gamma \omega_t.$$

Concentration evolves as

$$\omega_{t+1} = \rho \omega_t + \kappa H_t,$$

where H_t represents financial, political, informational, or technological concentration.

Contagion evolves as

$$c_{t+1} = \rho_c c_t + \eta_c \sigma_t.$$

6 Valuation Layer

Define

$$M_t = \begin{pmatrix} m_t \\ r_t \\ \lambda_t \end{pmatrix}.$$

Interpretation:

Variable	Interpretation
m_t	Pricing kernel state
r_t	Discount rate
λ_t	Risk aversion

The valuation kernel evolves as

$$m_{t+1} = m_t \exp(-r_t - \lambda_t \sigma_t).$$

Continuous time formulation:

$$dm_t = -m_t(r_t + \lambda_t \sigma_t)dt + m_t dZ_t.$$

7 Coupling Structure

The three subsystems interact.

Structural production influences valuation:

$$r_t = f(Y_t).$$

Volatility influences valuation:

$$\lambda_t = g(\sigma_t).$$

Valuation influences investment:

$$R_t = h(m_t).$$

The resulting feedback loop is

$$S_t \rightarrow V_t \rightarrow M_t \rightarrow S_t.$$

8 General Asset Pricing

For any asset with payoff stream

$$X_t,$$

the valuation equation is

$$P_t = \mathbb{E}_t \left[\sum_{s=t}^{\infty} M_{t,s} X_s \right].$$

This encompasses:

Asset	Payoff
Bond	Coupons, principal
Stock	Dividends
Firm	Free cash flow
State	Tax revenue
Alliance	Strategic utility
Weapon system	Military advantage
Technology platform	Future rents

9 State Prices and Geometry

Suppose

$$\Omega = \{\omega_1, \dots, \omega_n\}.$$

The valuation equation becomes

$$P_i = \sum_j M(\omega_j) \Pr(\omega_j) X_i(\omega_j).$$

Define state prices

$$q_j = M(\omega_j) \Pr(\omega_j).$$

Then

$$P_i = \langle q, X_i \rangle.$$

Asset pricing becomes a geometric projection problem.

10 Risk Premia

For gross return

$$R_i = \frac{X_i}{P_i},$$

we have

$$1 = \mathbb{E}[MR_i].$$

Expanding,

$$1 = \mathbb{E}[M]\mathbb{E}[R_i] + \text{Cov}(M, R_i).$$

Therefore

$$\boxed{\mathbb{E}[R_i] = \frac{1 - \text{Cov}(M, R_i)}{\mathbb{E}[M]}.$$

11 Political Economy

States possess future tax streams

$$T_t.$$

Their value is

$$V_{\text{state}} = \sum_t \mathbb{E}[M_t T_t].$$

Debt sustainability requires

$$\sum_t \mathbb{E}[M_t T_t] \geq D_0,$$

where D_0 is sovereign debt.

12 International Relations

Define geopolitical utility

$$GU_t.$$

Strategic valuation becomes

$$V_{\text{geo}} = \sum_t \mathbb{E}[M_t GU_t].$$

Military alliances, reserve currencies, logistics corridors, and technological ecosystems may all be represented as future utility streams.

13 Warfare Economics

Let

$$W_t$$

denote military advantage.

Weapon systems satisfy

$$V_{\text{weapon}} = \sum_t \mathbb{E}[M_t W_t].$$

Procurement becomes

$$\max \left(\sum_t \mathbb{E}[M_t W_t] - C_0 \right).$$

14 Civilizational Discounting

Define the effective discount rate

$$\tau_t = -\frac{\ln D(t, T)}{T - t}.$$

Definition 2. A civilization is long-horizon if

$$\tau_t$$

is persistently low.

Examples:

Low Discounting	High Discounting
Research	Extraction
Infrastructure	Consumption
Education	Patronage
Defense Modernization	Tactical Spending
Institution Building	Opportunism

15 Statistical Formulation

Observed prices satisfy

$$P_i = \mathbb{E}[MX_i] + \varepsilon_i.$$

The estimation problem is

$$\hat{\theta} = \arg \min_{\theta} \sum_i (P_i - \mathbb{E}[M_{\theta} X_i])^2.$$

The Fisher information matrix is

$$I(\theta) = \sum_i \left(\frac{\partial \mu_i}{\partial \theta} \right) \left(\frac{\partial \mu_i}{\partial \theta} \right)^T.$$

16 Machine Learning Interpretation

Let

$$M_\theta = N_\theta(Z_t)$$

be a neural network.

Inputs:

$$Z_t = (K_t, L_t, A_t, I_t, G_t, \sigma_t, \omega_t, c_t).$$

Training objective:

$$\min_{\theta} \sum_i (P_i - \mathbb{E}[M_\theta X_i])^2.$$

Markets become sensors of hidden civilization dynamics.

17 The Inverse Problem

Suppose observed variables are

$$Y_t^{obs}.$$

The observation equation is

$$Y_t^{obs} = H(X_t) + \varepsilon_t.$$

State dynamics satisfy

$$X_{t+1} = F(X_t, \eta_t).$$

The objective becomes

$$\boxed{X_t = \Pi^{-1}(Y_t^{obs}).}$$

18 Main Theorem

Theorem 1. *Suppose*

$$X_t = (S_t, V_t, M_t)$$

is a latent civilization state generating all observable prices through

$$P_i = \mathbb{E}[M X_i].$$

*Then every asset price is a projection of the civilization state onto the payoff space.
Consequently asset pricing may be formulated as a latent-state estimation problem.*

Proof. Since

$$P_i = \mathbb{E}[M X_i],$$

all prices depend on the valuation component M .

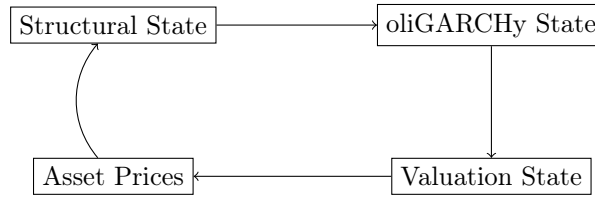
Since M itself depends on structural and volatility variables through the coupling equations,

$$P_i = \Pi(S_t, V_t, M_t).$$

Hence observed prices are functions of the civilization state.

The inverse problem reconstructs the hidden state from these observables. □

19 A Civilization-State Diagram



20 Conclusion

We have developed a unified framework linking Structural Economics, oliGARCHy dynamics, stochastic discount factors, asset pricing, political economy, international relations, warfare economics, statistics, and machine learning.

The central insight is that asset prices are not primitive objects.

Rather,

Asset Prices = Observable Projections of a Hidden Civilization State.

The stochastic discount factor serves as the bridge connecting production, uncertainty, valuation, and strategic behavior.

The resulting theory transforms finance into a general science of valuation and civilization-state inference.

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Glossary

Civilization State

Latent vector containing structural, volatility, and valuation states.

Structural Economics

Theory governing production, institutions, and technology.

oliGARCHy

Framework describing endogenous volatility, concentration, and contagion.

SDF

Stochastic Discount Factor.

Pricing Kernel

Alternative name for the SDF.

State Price

Price of a state-contingent payoff.

Geopolitical Utility

Future strategic benefit.

Civilizational Discounting

Intertemporal preference at societal scale.

Latent-State Estimation

Recovery of hidden states from observables.

General Valuation Theory

Unified framework for pricing economic, political, and strategic assets.

The End